SCREW SEPARATOR COOLING JACKET - DESIGN OPTIMIZATION SUMMARY

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FIXED OPERATING CONDITIONS:

  Coolant mass flow rate:    0.25 kg/s (Helisol 5A)
  Screw conveyor length:     1.2 m
  Separator diameter:        101.6 mm
  Coolant inlet temperature: 281 C
  Wall inlet temperature:    500 C
  Wall outlet temperature:   285 C
  Required heat extraction:  1100 W
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DESIGN COMPARISON

Parameter	Original	V1 (Balanced)	V2 (Min DP)	Units
GAP (Annular Gap)	10.0	22.0	25.0	mm
w (Channel Width)	30.0	55.0	65.0	mm
$t_w$all (Wall Thick.)	20.0	15.0	18.0	mm
Pitch (w + $t_w$all)	50.0	70.0	83.0	mm
Pressure Drop	36.38	0.96	0.41	kPa
DP Reduction	-	97.4	98.9	%
Heat Transfer Coeff	1288.6	375.7	289.7	W/(m2K)
Reynolds Number	30488	15838	13550	-
Velocity	1.21	0.30	0.22	m/s
Number of Turns	24.0	17.1	14.5	-
Coil Length	7.75	5.50	4.66	m
Safety Factor (A_av/$A_r$eq)	23.9	7.0	5.4	-

DESIGN DESCRIPTIONS:

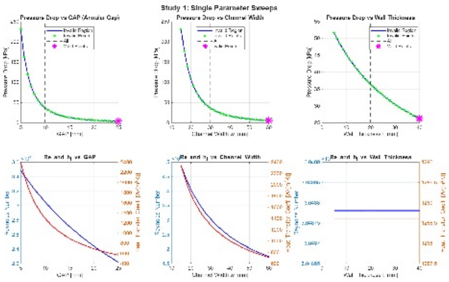
Original Baseline: Starting design with GAP=10mm, w=30mm, $t_w$all=20mm. Pressure drop at 36.38 kPa. Standard helical cooling jacket configuration.

V1 (Balanced Design): Optimized to reduce pressure drop while maintaining good heat transfer. GAP=22mm, w=55mm, $t_w$all=15mm. Achieved 97.4% DP reduction with safety factor of 7.0x.

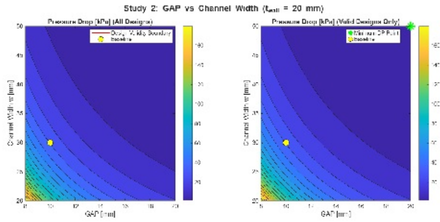
V2 (Minimum Pressure Drop): Optimized primarily to minimize pressure drop. GAP=25mm, w=65mm, $t_w$all=18mm. Larger channels and annular gap. Achieved 98.9% DP reduction (36.38 kPa -> 0.41 kPa).

STUDY 1: SINGLE PARAMETER SWEEPS & INITIAL OPTIMIZATION

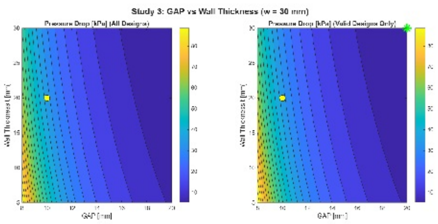
Single Parameter Sweeps



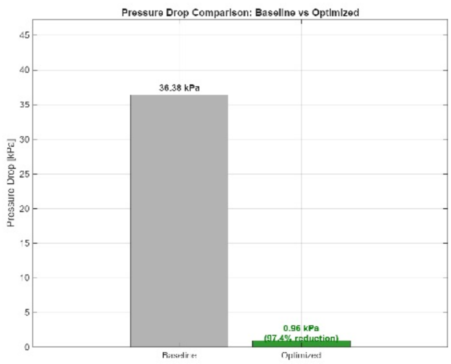
GAP vs Channel Width Contour



GAP vs Wall Thickness Contour



Baseline vs Optimized



KEY FINDINGS - STUDY 1:

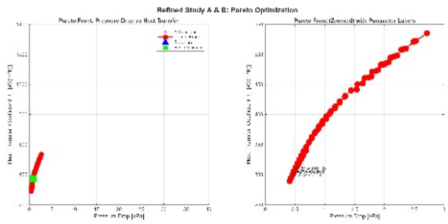
GAP is the dominant parameter - increasing from 10mm to 22mm reduced DP by 97%

Pressure drop scales with velocity squared (v^2) - larger channels dramatically reduce losses

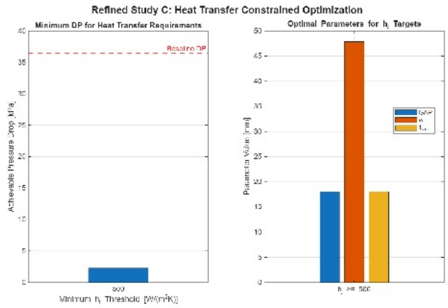
Trade-off: Higher GAP and w reduce heat transfer coefficient (h_j) but design remains valid

STUDY 2: REFINED OPTIMIZATION & PARETO ANALYSIS

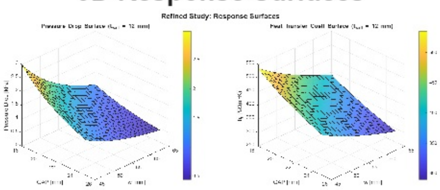
Pareto Front: DP vs h_j Trade-off



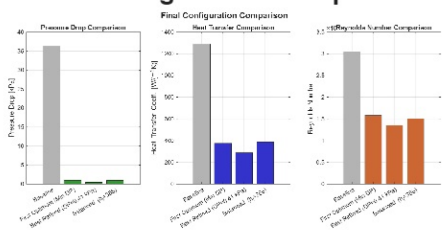
h_j Constrained Optimization



3D Response Surfaces



Multi-Configuration Comparison



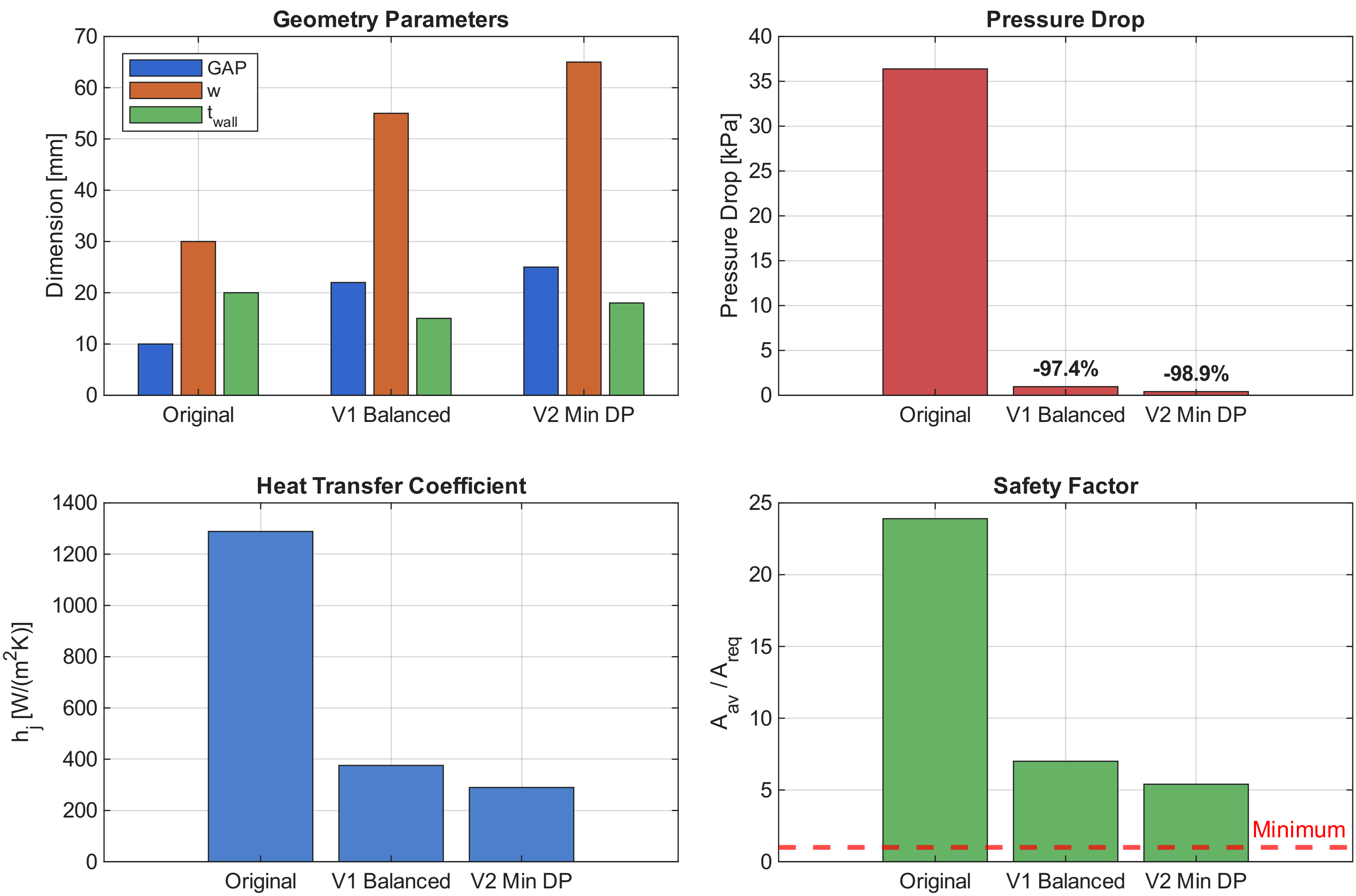
KEY FINDINGS - STUDY 2 (REFINED):

Pareto front shows clear trade-off between pressure drop and heat transfer coefficient

Minimum DP of 0.41 kPa achieved at GAP=25mm, w=65mm, $t_w$all=18mm

All Pareto-optimal designs maintain safety factor > 5x (design valid)

DESIGN RECOMMENDATIONS



RECOMMENDATIONS:

Choose V1 BALANCED DESIGN (Recommended) if:
- Good balance between low pressure drop and heat transfer is needed
- Comfortable safety margin (7x) is preferred
- Moderate jacket dimensions are acceptable

Choose V2 MIN DP DESIGN if:
- Minimizing pumping power is the primary concern
- System can accommodate larger jacket dimensions
- Lower heat transfer coefficient (289.7 W/m2K) is acceptable

AVOID ORIGINAL DESIGN:
- Excessively high pressure drop (36.38 kPa)
- No benefit from oversized heat transfer coefficient

All optimized designs meet thermal requirements ($Q_r$eq = 1100 W) with safety factors well above 1.0, ensuring robust heat transfer performance.